Modern Portfolio Theory as a Vector Calculus Framework: Risk Aversion in Northwestern Investment Management Group's Portfolio

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Abstract

This project develops the theory of mean-variance optimization from first principles, beginning with a two-asset portfolio and extending the analysis to a generalized n-asset portfolio. To connect portfolio theory with vector calculus, scalar line integrals are introduced and used to compute the average utility across all feasible allocations of a two-asset portfolio. This framework is then applied to the Northwestern Investment Management Group's portfolio (the Portfolio). We reverse-optimize the Portfolio Committee's implied risk aversion constant and mean-variance optimize the Portfolio.

Contents

1 Introduction

While identifying assets with attractive expected returns is a fundamental task in investment management, constructing an optimal portfolio from those assets is equally important. Maximizing the expected return alone would lead an investor to concentrate the entire portfolio in a single asset, exposing the portfolio to substantial asset-specific risk should that single asset perform poorly. Portfolio construction, accordingly, requires an investor to consider not only the expected return but also how assets interact with each other. This interaction is best captured by covariance, which measures the extent to which asset returns move together. The trade-off between expected return and covariance-driven risk sits at the heart of modern portfolio theory and mean-variance optimization.

This project seeks to (i) develop the mathematics behind mean-variance optimization from first principles, (ii) apply mean-variance optimization to Northwestern Investment Management Group's portfolio, and (iii) connect the framework to several core ideas from vector calculus and linear algebra.

2 Mean-Variance Optimization

In this section, we develop the theory of mean-variance optimization by first examining a two-asset portfolio and then extending this analysis to a generalized n-asset portfolio.

2.1 Two-Asset Portfolio

Setup. Let us consider a simple portfolio with two assets x_1, x_2 each with projected returns μ_1, μ_2 respectively. Let x be the asset vector, R the actual return vector, μ the expected return vector, w the portfolio weight vector, and σ the standard deviation vector, such that

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \text{Stock 1} \\ \text{Stock 2} \end{bmatrix}$$

describes the portfolio with two stocks: Stock 1 and Stock 2.

Expected Return. Then, the actual return of the portfolio x is

$$R_x = w_1 R_1 + w_2 R_2$$

such that the expected return of the portfolio x is

$$\mu_x = E[R_x] = w_1\mu_1 + w_2\mu_2$$

Because the portfolio x has only two assets, $w_1 + w_2 = 1$ must hold, and thus $w_2 = 1 - w_1$ such that

$$E[R_x] = w_1\mu_1 + (1 - w_1)\mu_2$$

Portfolio Variance. Now, let us consider the portfolio variance of x,

$$\text{Var}(R_x) = \text{Var}(w_1 R_1 + (1 - w_1)R_2)$$

Recall

$$\text{Var}(X) = E[(X - E[X])^2]$$

and let $X = w_1 R_1 + (1 - w_1)R_2$, such that

$$\begin{aligned}
\text{Var}(R_x) &= E[(w_1(R_1 - \mu_1) + (1 - w_1)(R_2 - \mu_2))^2] \\
&= E[w_1^2(R_1 - \mu_1)^2 + (1 - w_1)^2(R_2 - \mu_2)^2 + 2w_1(1 - w_1)(R_1 - \mu_1)(R_2 - \mu_2)] \\
&= w_1^2 E[(R_1 - \mu_1)^2] + (1 - w_1)^2 E[(R_2 - \mu_2)^2] + 2w_1(1 - w_1)E[(R_1 - \mu_1)(R_2 - \mu_2)]
\end{aligned}$$

Since $E[(R_1 - \mu_1)^2] = \sigma_1^2, E[(R_2 - \mu_2)^2] = \sigma_2^2, E[(R_1 - \mu_1)(R_2 - \mu_2)] = \text{Cov}(R_1, R_2) = \sigma_{12}$,

$$\text{Var}(R_x) = w_1^2\sigma_1^2 + (1 - w_1)^2\sigma_2^2 + 2w_1(1 - w_1)\sigma_{12}$$

Objective. Having deduced the equations for expected return and portfolio variance of x, we can construct the following utility optimization problem:

$$\max_{w_1} U(w_1) = E[R_x] - \frac{\gamma}{2}\sigma_x^2$$

where γ represents the risk aversion constant.

Substituting the equations into the utility function gives:

$$U(w_1) = w_1\mu_1 + (1 - w_1)\mu_2 - \frac{\gamma}{2}[w_1^2\sigma_1^2 + (1 - w_1)^2\sigma_2^2 + 2w_1(1 - w_1)\sigma_{12}]$$

Taking the derivative of both sides with respect to w_1 and setting it equal to zero for the first-order condition gives:

$$\frac{dU}{dw_1} = \mu_1 - \mu_2 - \gamma[w_1\sigma_1^2 - (1 - w_1)\sigma_2^2 + (1 - 2w_1)\sigma_{12}] = 0$$

Optimal Portfolio. Solving for w_1 gives:

$$\boxed{w_1^* = \frac{\frac{\mu_1 - \mu_2}{\gamma} + \sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}}$$

Before extending this analysis to a generalized n-asset portfolio, we apply the framework to an actual two-asset portfolio and verify the effectiveness of mean-variance optimization. To do so, we compare the utility value of the mean-variance optimized portfolio (MVO), $U(w_1)_{\text{MVO}}$, with the average utility across all possible portfolio allocations.

2.2 Example of Two-Asset Portfolio: STZ and WM

2.2.1 Mean-Variance Optimized Two-Asset Portfolio

Setup. Consider a two-asset portfolio consisting of two holdings currently included in Northwestern Investment Management Group's portfolio: Constellation Brands (STZ) and Waste Management (WM).

For STZ, IMG estimates bear-, base-, and bull-case price targets of $204.87, $224.72, and $251.93, respectively. Assuming a probability distribution of 25%, 50%, and 25% across the three scenarios, the weighted price target is $226.56. Given the current share price of $138.82 (as of May 30, 2026), this implies an expected return of 63%.

For WM, IMG estimates bear-, base-, and bull-case price targets of $239.89, $319.39, and $389.89, respectively. Assuming a probability distribution of 20%, 50%, and 30% across the three scenarios, the weighted price target is $324.64. Given the current share price of $211.46 (as of May 30, 2026), this implies an expected return of 54%.

In addition to the expected returns, the mean–variance optimization framework requires the variances of the individual assets, the covariance between them, and a risk aversion constant as inputs. For the risk aversion constant, let us assume $\gamma = 25$, since the values of γ between 1 and 50 are commonly considered in the literature. For the variance of STZ and WM, and the covariance between the two, we utilized Microsoft Excel's built-in functions over a monthly return for each stock over the previous three years.

We define:

- $x = (\text{STZ}, \text{WM})^\top$
- $\mu = (0.63, 0.54)^\top$
- $\gamma = 25$
- $\sigma_1^2 = 0.00483, \sigma_2^2 = 0.00307, \sigma_{12} = 0.00064$

Expected Return. Then, the expected return of this portfolio is:

$$\mu = E[R] = 0.63w_1 + 0.54w_2 = 0.63w_1 + 0.54(1 - w_1)$$

Portfolio Variance. And the portfolio variance is:

$$\text{Var}(R) = w_1^2(0.00483) + (1 - w_1)^2(0.00307) + 2w_1(1 - w_1)(0.00064)$$

Objective. Accordingly, our utility function is:

$$U(w_1) = 0.63w_1 + 0.54(1 - w_1) - \frac{25}{2}[w_1^2(0.00483) + (1 - w_1)^2(0.00307) + 2w_1(1 - w_1)(0.00064)]$$

Taking the derivative of both sides with respect to w_1 and setting it equal to zero for the first-order condition gives:

$$\frac{dU}{dw_1} = 0.63 - 0.54 - 25[w_1(0.00483) - (1 - w_1)(0.00307) + (1 - 2w_1)(0.00064)] = 0$$

Optimal Portfolio. Solving for w_1 gives:

$$w_1^* = \frac{\frac{0.63-0.54}{25} + 0.00307 - 0.00064}{0.00483 + 0.00307 - 2(0.00064)}$$

$$w_1^* = \frac{0.00603}{0.00662}$$

$$\boxed{w_1^* = 0.91}$$

This result implies that, in order to optimize this two-asset portfolio, we should allocate 91% of the portfolio to STZ and 9% to WM. Then, the utility value of the mean-variance optimized portfolio is:

$$U(0.91)_{\text{MVO}} = 0.63(0.91) + 0.54(0.09) - \frac{25}{2}[(0.91)^2(0.00483) + (0.09)^2(0.00307) + 2(0.91)(0.09)(0.00064)]$$

$$U(0.91)_{\text{MVO}} = 0.5733 + 0.0486 - 12.5[0.00399 + 0.00002 + 0.0001]$$

$$\boxed{U(0.91)_{\text{MVO}} = 0.57053}$$

In other words, the mean-variance optimized portfolio has a risk-adjusted expected return of 57%.

2.2.2 Scalar Line Integral

To determine the average utility across all possible portfolio allocations, we represent each feasible portfolio by its corresponding weight pair (w_1, w_2), where $w_1 + w_2 = 1$. Then, the projected utility of a given portfolio allocation is expressed as a function $U(w_1, w_2)$, which can be considered as a surface above the (w_1, w_2)-plane. The average utility across all feasible allocations can then be calculated by a scalar line integral of $U(w_1, w_2)$ over the feasible set $w_1 + w_2 = 1$, divided by the length of the line. Before we proceed with this calculation, however, let us introduce scalar line integral.

Conceptually, scalar line integral adds up the value of a scalar function along a path. Let

$$w : [a, b] \to R^2$$

be a path of class C^1, which means that $w(t)$ yields a point in R^2 for all values of $t \in [a, b]$ and that the path is differentiable with continuous derivative. In our setting, $w(t)$ represents a portfolio allocation at a parameter value t.

Let

$$U : W \subseteq R^2 \to R$$

be a continuous scalar function whose domain W contains the image of w, such that the composite function $U(w(t))$ is well-defined. In our setting, U assigns a utility value, or the risk-adjusted expected return, to all possible portfolio allocations.

In order to introduce the scalar line integral from first principles, let

$$a = t_0 < t_1 < \cdots < t_k < \cdots < t_n = b$$

be a partition of $[a, b]$ such that each subinterval $[t_{k-1}, t_k]$ corresponds to a small segment of the path and $\Delta t_k = t_k - t_{k-1}$.

Then, for a point $t_k^* \in [t_{k-1}, t_k]$, the corresponding point on the curve is:

$$w(t_k^*)$$

And at this point $w(t_k^*)$, the scalar function has value,

$$U(w(t_k^*))$$

which is the utility of the portfolio, whose allocation is represented by $w(t_k^*)$.

In order to compute the length of the segment $[t_{k-1}, t_k]$, we recall that:

$$distance = speed \times time$$

Since $w'(t)$ is the velocity vector of the path by definition, its magnitude

$$||w'(t)||$$

is the speed. Then, the corresponding time period of travel is Δt_k, such that over a small segment of the path $[t_{k-1}, t_k]$, the distance traveled along the path is:

$$||w'(t)||\Delta t_k$$

Then, the sum of the utility value over this segment is:

$$U(w(t_k^*))||w'(t_k^*)||\Delta t_k$$

such that summing this value across the entire path yields the following Riemann sum:

$$\sum_{k=1}^{n} U(w(t_k^*))||w'(t_k^*)||\Delta t_k$$

Now, as is the case with all integrals, the scalar line integral is a limit of appropriate Riemann sum:

$$\int_a^b U(w(t))||w'(t)||dt = \lim_{\text{all } \Delta t_k \to 0} \sum_{k=1}^{n} U(w(t_k^*))||w'(t_k^*)||\Delta t_k$$

Definition 1. Therefore, the scalar line integral of U along the C^1 path w is

$$\boxed{\int_w U ds = \int_a^b U(w(t))||w'(t)||dt}$$

Having introduced the definition of scalar line integral, let us consider the following example before applying it to our two-asset portfolio.

2.2.3 A Fully Worked Example of Scalar Line Integral

Setup. Suppose a hiker is walking a mountain trail and they want to know the accumulated elevation along the trail. Suppose the following function represents the elevation at each point in the xy-plane in meters:

$$h(x, y) = 100 + 2x + y$$

and that the path of the mountain trail is:

$$r(t) = (t, t^2), \qquad 0 \leq t \leq 2$$

Then,

$$h(r(t)) = 100 + 2t + t^2$$

Scalar Line Integral. Recall the definition of scalar line integral:

$$\int_r h ds = \int_0^2 h(r(t))||r'(t)||dt$$

First,

$$r'(t) = (1, 2t)$$

So,

$$||r'(t)|| = \sqrt{1^2 + (2t)^2} = \sqrt{1 + 4t^2}$$

such that

$$\int_r h ds = \int_0^2 (100 + 2t + t^2)(\sqrt{1 + 4t^2})dt$$

$$= 100 \int_0^2 \sqrt{1 + 4t^2}dt + 2 \int_0^2 t\sqrt{1 + 4t^2}dt + \int_0^2 t^2\sqrt{1 + 4t^2}dt$$

Let I_1, I_2, and I_3 represent the three integrals above, respectively, such that

$$I_1 = 100\int_0^2 \sqrt{1+4t^2}dt, I_2 = 2\int_0^2 t\sqrt{1+4t^2}dt, I_3 = \int_0^2 t^2\sqrt{1+4t^2}dt$$

Integral 1. For I_1, let $2t = \tan\theta$, such that

$$\frac{dt}{d\theta} = \frac{1}{2}\sec^2\theta$$

$$t = 2 \implies \frac{1}{2}\tan\theta = 2 \implies \theta = \arctan(4)$$

$$t = 0 \implies \frac{1}{2}\tan\theta = 0 \implies \theta = \arctan(0) = 0$$

Then,

$$I_1 = 100\int_0^{\arctan(4)} \sqrt{1+\tan^2\theta}dt = 100\int_0^{\arctan(4)} \sec\theta(\frac{1}{2}\sec^2\theta)d\theta = 50\int_0^{\arctan(4)} \sec\theta\sec^2\theta d\theta$$

Use integration by parts:

$$u = \sec\theta, dv = \sec^2\theta d\theta$$

$$du = \sec\theta\tan\theta d\theta, v = \tan\theta$$

such that,

$$I_1 = 50([\sec\theta\tan\theta]_0^{\arctan(4)} - \int_0^{\arctan(4)} \tan\theta(\sec\theta\tan\theta)d\theta)$$

$$= 50(\sec(\arctan(4))\tan(\arctan(4)) - \sec(0)\tan(0) - \int_0^{\arctan(4)} \sec\theta\tan^2\theta d\theta)$$

$$= 50(4\sec(\arctan(4)) - \int_0^{\arctan(4)} \sec\theta(\sec^2\theta - 1)d\theta)$$

Note the following:

$$I_1 = 50\int_0^{\arctan(4)} \sec^3\theta d\theta = 200\sec(\arctan(4)) - 50\int_0^{\arctan(4)} \sec^3\theta d\theta + 50\int_0^{\arctan(4)} \sec\theta d\theta$$

$$2I_1 = 200\sec(\arctan(4)) + 50\int_0^{\arctan(4)} \sec\theta d\theta$$

Let $u = \sec\theta + \tan\theta$, such that

$$\frac{du}{d\theta} = \sec\theta\tan\theta + \sec^2\theta$$

$$\theta = \arctan(4) \implies u = \sec(\arctan(4)) + \tan(\arctan(4)) \implies u = \sqrt{17} + 4$$

$$\theta = 0 \implies u = \sec(\arctan(0)) + \tan(\arctan(0)) \implies u = 1$$

Then,

$$I_1 = 100\sec(\arctan(4)) + 25\int_1^{\sqrt{17}+4} \frac{\sec\theta(\sec\theta+\tan\theta)}{\sec\theta+\tan\theta}d\theta$$

$$= 100\sec(\arctan(4)) + 25\int_1^{\sqrt{17}+4} \frac{\sec^2\theta+\sec\theta\tan\theta}{\sec\theta+\tan\theta}(\frac{1}{\sec\theta\tan\theta+\sec^2\theta})du$$

$$= 100\sec(\arctan(4)) + 25\int_1^{\sqrt{17}+4} \frac{1}{u}du$$

$$= 100\sqrt{17} + 25[\ln|u|]_1^{\sqrt{17}+4}$$

$$= 100\sqrt{17} + 25[\ln(\sqrt{17}+4) - \ln(1)]$$

Therefore,

$$\boxed{I_1 = 100\sqrt{17} + 25\ln(\sqrt{17}+4)}$$

Integral 2. For I_2, let $u = 1 + 4t^2$, such that

$$\frac{du}{dt} = 8t$$

$$t = 2 \implies u = 1 + 4(2)^2 \implies u = 17$$

$$t = 0 \implies u = 1 + 4(0)^2 \implies u = 1$$

Then,

$$I_2 = 2\int_1^{17} t\sqrt{u}(\frac{1}{8t})du$$

$$= \frac{1}{4}\int_1^{17} \sqrt{u}du$$

$$= \frac{1}{4}[\frac{2}{3}u^{\frac{3}{2}}]_1^{17}$$

$$= \frac{1}{6}(17^{\frac{3}{2}} - 1)$$

Therefore,

$$\boxed{I_2 = \frac{1}{6}(17\sqrt{17} - 1)}$$

Integral 3. For I_3, let $u = 2t$, such that

$$\frac{du}{dt} = 2$$

$$t = 2 \implies u = 2(2) \implies u = 4$$

$$t = 0 \implies u = 2(0) \implies u = 0$$

Then,

$$I_3 = \int_0^4 t^2\sqrt{1+u^2}(\frac{1}{2})du$$
$$= \frac{1}{8}\int_0^4 u^2\sqrt{1+u^2}du$$

Let $u = \tan\theta$, such that

$$\frac{du}{d\theta} = \sec^2\theta$$
$$u = 4 \implies \theta = \arctan(4)$$
$$u = 0 \implies \theta = \arctan(0) = 0$$

Then,

$$I_3 = \frac{1}{8}\int_0^{\arctan 4} \tan^2\theta\sec\theta(\sec^2\theta)d\theta$$
$$= \frac{1}{8}\int_0^{\arctan(4)} \tan^2\theta\sec^3\theta d\theta$$

Since $\tan^2\theta = \sec^2\theta - 1$,

$$I_3 = \frac{1}{8}\int_0^{\arctan(4)} (\sec^2\theta - 1)\sec^3\theta d\theta$$
$$= \frac{1}{8}\int_0^{\arctan(4)} \sec^5\theta d\theta - \frac{1}{8}\int_0^{\arctan(4)} \sec^3\theta d\theta$$
$$= \frac{1}{8}[\frac{1}{4}[\sec^3\theta\tan\theta]_0^{\arctan(4)} + \frac{3}{4}\int_0^{\arctan(4)} \sec^3\theta d\theta - \int_0^{\arctan(4)} \sec^3\theta d\theta]$$
$$= \frac{1}{32}(\sec^3(\arctan(4))\tan(\arctan(4) - \sec^3(0)\tan(0) - \int_0^{\arctan(4)} \sec^3\theta d\theta)$$
$$= \frac{1}{32}(68\sqrt{17} - \int_0^{\arctan(4)} \sec^3\theta d\theta)$$
$$= \frac{17\sqrt{17}}{8} - \frac{1}{32}\int_0^{\arctan(4)} \sec^3\theta d\theta$$

Substituting the $\sec^3\theta$ formula we have derived when solving for I_1 gives:

$$I_3 = \frac{17\sqrt{17}}{8} - \frac{1}{32}[\frac{1}{2}\sec\theta\tan\theta + \frac{1}{2}\ln(\sec\theta + \tan\theta)]_0^{\arctan(4)}$$
$$= \frac{17\sqrt{17}}{8} - \frac{1}{64}(4\sec(\arctan(4)) + \ln(\sec(\arctan(4)) + 4) - \sec 0\tan 0 - \ln(\sec 0 + \tan 0))$$
$$= \frac{17\sqrt{17}}{8} - \frac{1}{64}(4\sqrt{17} + \ln(\sqrt{17} + 4))$$

Therefore,

$$\boxed{I_3 = \frac{17\sqrt{17}}{8} - \frac{\sqrt{17}}{16} - \frac{\ln(\sqrt{17}+4)}{64}}$$

Summing the three integrals,

$$\int_r h ds = I_1 + I_2 + I_3$$

$$\int_r h ds = 100\sqrt{17} + 25\ln(\sqrt{17}+4) + \frac{17\sqrt{17}}{6} - \frac{1}{6} + \frac{17\sqrt{17}}{8} - \frac{\sqrt{17}}{16} - \frac{\ln(\sqrt{17}+4)}{64}$$

$$\int_r h ds = \frac{5035\sqrt{17}}{48} - \frac{1}{6} + \frac{1599}{64}\ln(4+\sqrt{17}) \approx 484.67$$

Therefore, the scalar line integral evaluates to:

$$\boxed{\int_r h ds \approx 484.67}$$

which represents the accumulated value of the elevation function along the mountain trail:

$$r(t) = (t, t^2), \qquad 0 \leq t \leq 2$$

2.2.4 Average Utility across All Possible Two-Asset Portfolio Allocations

Setup. To compute the average utility across all possible portfolio allocations for our two-asset portfolio, we first calculate the total utility accumulated over the feasible set $w_1 + w_2 = 1$ through scalar line integral. Then, we divide this sum by the length of the feasible set.

Recall the utility function of our two-asset portfolio from Section 2.2.1:

$$U(w_1, w_2) = 0.63w_1 + 0.54w_2 - \frac{25}{2}[w_1^2(0.00483) + w_2^2(0.00307) + 2w_1w_2(0.00064)]$$

and the path of our feasible set:

$$w(w_1) = (w_1, 1-w_1), \qquad 0 \leq w_1 \leq 1$$

Then,

$$U(w(w_1)) = 0.63w_1 + 0.54(1-w_1) - \frac{25}{2}[w_1^2(0.00483) + (1-w_1)^2(0.00307) + 2w_1(1-w_1)(0.00064)]$$

Recall the definition of scalar line integral from Section 2.2.2,

$$\int_w U ds = \int_0^1 U(w(w_1))||w'(w_1)||dw_1$$

First,

$$w'(w_1) = (1, -1)$$

So,

$$||w'(w_1)|| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

such that

$$\int_w U ds = \int_0^1 0.63w_1 + 0.54(1-w_1) - \frac{25}{2}[w_1^2(0.00483) + (1-w_1)^2(0.00307) + 2w_1(1-w_1)(0.00064)](\sqrt{2})dw_1$$

$$= \sqrt{2}\int_0^1 (-0.08275w_1^2 + 0.15075w_1 + 0.501625)dw_1$$

$$= -0.08275\sqrt{2}\int_0^1 w_1^2 dw_1 + 0.15075\sqrt{2}\int_0^1 w_1 dw_1 + 0.501625\sqrt{2}\int_0^1 1dw_1$$

$$= -0.08275\sqrt{2}[\frac{w_1^3}{3}]_0^1 + 0.15075\sqrt{2}[\frac{w_1^2}{2}]_0^1 + 0.501625\sqrt{2}[w_1]_0^1$$

$$= -0.08275\sqrt{2}(\frac{1}{3}) + 0.15075\sqrt{2}(\frac{1}{2}) + 0.501625\sqrt{2}$$

$$= 0.54942\sqrt{2} \approx 0.77699$$

Therefore, the scalar line integral evaluates to:

$$\boxed{\int_w U ds \approx 0.77699}$$

which represents the accumulated utility across all possible portfolio allocations along our feasible path:

$$w(w_1) = (w_1, 1-w_1), \qquad 0 \leq w_1 \leq 1$$

To compute the average utility across all possible portfolio allocations, we divide this value by the length of the path $w(w_1) = (w_1, 1-w_1), 0 \leq w_1 \leq 1$, which equals:

$$L = \sqrt{(0-1)^2 + (1-0)^2} = \sqrt{2}$$

Therefore, the average utility across all possible portfolio allocations of the two-asset portfolio equals:

$$\boxed{\frac{1}{L}\int_w U ds = \frac{1}{\sqrt{2}}(0.54942\sqrt{2}) = 0.54942}$$

Then, we can conclude that the average risk-adjusted expected return across all possible portfolio allocations is 55%, while the mean-variance optimized portfolio from Section 2.2.1 has a risk-adjusted expected return of 57%. Therefore, we find that the mean-variance optimized portfolio achieves a higher risk-adjusted expected return than the average feasible portfolio allocation, supporting the framework's effectiveness.

Having applied mean-variance optimization to a two-asset portfolio example and verified the effectiveness of mean-variance optimization, we now extend the analysis of mean-variance optimization to a generalized n-asset portfolio.

2.3 n-Asset Portfolio

Mean-variance optimization of a two-asset portfolio in Section 2.1 solved for a single weight, w_1, using a simplified formula. In this section, we repeat the same logic for an n-asset portfolio to derive a general formula, which can be applied to Northwestern Investment Management Group's 14-asset portfolio.

Setup. For an n-asset portfolio, we define:

- $w = (w_1, \ldots, w_n)^\top$, the weight vector, where w_i is the percentage of the portfolio invested in asset i;
- $\mu = (\mu_1, \ldots, \mu_n)^\top$, the expected return vector, where $\mu_i = E[R_i]$;
- Σ, the $n \times n$ covariance matrix, with entries $\Sigma_{ij} = \mathrm{Cov}(R_i, R_j)$ and diagonal $\Sigma_{ii} = \sigma_i^2$;
- $\gamma > 0$, the risk-aversion constant.

Expected Return. The expected portfolio return is the weighted sum of the individual expected returns:

$$E[R_P] = \sum_{i=1}^{n} w_i \mu_i = w^\top \mu$$

Portfolio Variance. The portfolio variance sums the contribution of every pair of assets, which in matrix form is the quadratic form:

$$\mathrm{Var}(R_P) = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \, \Sigma_{ij} = w^\top \Sigma w$$

Objective. As in the two-asset case, the investor seeks to maximize the risk-adjusted expected return, which is represented by the following utility optimization problem:

$$\max_{w} \; U(w) = w^\top \mu - \frac{\gamma}{2} \, w^\top \Sigma w$$

With many weights, we differentiate using the gradient ∇_w, the vector of partial derivatives with respect to each weight. Two standard identities apply:

$$\nabla_w (w^\top \mu) = \mu, \qquad \nabla_w (w^\top \Sigma w) = 2\Sigma w \quad (\Sigma \text{ symmetric}).$$

Taking the derivative of both sides of the optimization problem with respect to the weight vector w and setting it equal to zero for the first-order condition gives:

$$\nabla_w U(w) = \mu - \gamma \Sigma w = 0.$$

Optimal Portfolio. Then,

$$\gamma \Sigma w = \mu,$$

Assuming that the covariance matrix Σ is invertible[1], which is equivalent to assuming that no asset's return can be expressed as a linear combination of the others, we may multiply both sides by Σ^{-1} to obtain[2]:

$$\boxed{w^* = \frac{1}{\gamma} \Sigma^{-1} \mu}$$

Having extended our analysis of mean-variance optimization to a generalized n-asset portfolio, we now apply this framework to Northwestern Investment Management Group's portfolio (the Portfolio).

3 Application of Mean-Variance Optimization

Recall from Section 2.2.1 that the mean-variance optimization requires the expected returns, the covariance matrix, and a risk aversion constant as inputs. While the expected returns can be taken from the Northwestern Investment Management Group's internal valuation models, the covariance matrix and the risk aversion constant have to be computed.

3.1 Building the Covariance Matrix from Price Data

The covariance matrix must be estimated from price data, since it has to be estimated from how the stocks have actually behaved in the past. We built it from eight months of monthly closing prices for each holding, spanning October 2025 through May 2026, following four steps.[3]

Step 1: From Prices to Returns. Since a covariance matrix describes how *returns* move, not how prices move, the first step is to convert each stock's price history into a return history. For a stock with closing price P_t at the end of month t, the monthly return is the percentage change from the previous month,

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

The eight months of prices give nine price observations per stock, and therefore eight monthly returns per stock.

[1] This is a standard assumption in modern portfolio theory since it is highly unlikely for an asset's return to be a linear combination of other assets' returns.

[2] While this equation does not explicitly include the budget constraint, $\sum_{i=1}^{n} w_i = 1$, the Python optimization used in our application of mean-variance optimization incorporates this constraint, ensuring that the portfolio weights sum to 100%.

[3] We chose eight months since the Portfolio includes some stocks that completed its Initial Public Offering only eight months ago.

Step 2: The Sample Mean. For each stock i, we compute the average of its monthly returns over the window. This is the *sample mean return*, our best estimate of the stock's typical monthly move:

$$\bar{R}_i = \frac{1}{T}\sum_{t=1}^{T} R_{i,t}$$

where T is the number of monthly returns and $R_{i,t}$ is stock i's return in month t.

Step 3: Variances and Covariances. With the means in hand, we measure how each stock varies around its own mean, and how each pair of stocks varies together. The two following formulae measure sample variance and sample covariance:

$$\hat{\sigma}_i^2 = \frac{1}{T-1}\sum_{t=1}^{T}\big(R_{i,t}-\bar{R}_i\big)^2, \qquad \hat{\sigma}_{ij} = \frac{1}{T-1}\sum_{t=1}^{T}\big(R_{i,t}-\bar{R}_i\big)\big(R_{j,t}-\bar{R}_j\big).$$

The variance $\hat{\sigma}_i^2$ measures stock i's own risk; The covariance $\hat{\sigma}_{ij}$ is positive when the two stocks tend to rise and fall together, and negative when they tend to move oppositely. We divide by $T-1$ rather than T for a technical reason known as Bessel's correction: Because we have already used the data to estimate the means $\bar{R}_i$, dividing by $T-1$ removes a small downward bias and makes the estimate accurate on average. We note that with only eight monthly returns and twelve to fourteen assets, the estimate rests on far fewer observations than there are entries to estimate. The resulting matrix is, therefore, noisy and this estimation error is a central weakness of the framework rather than a detail of our particular sample.

Step 4: Assembling the Matrix. Finally we arrange these numbers into the $n \times n$ covariance matrix Σ, placing the variances on the diagonal and the covariances off it:

$$\Sigma = \begin{bmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{1,2} & \cdots & \hat{\sigma}_{1,n} \\ \hat{\sigma}_{2,1} & \hat{\sigma}_2^2 & \cdots & \hat{\sigma}_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\sigma}_{n,1} & \hat{\sigma}_{n,2} & \cdots & \hat{\sigma}_n^2 \end{bmatrix}.$$

Because $\hat{\sigma}_{ij} = \hat{\sigma}_{ji}$ (i.e. the covariance of i with j is the same as j with i), the matrix is symmetric; It is a mirror image across its diagonal. We carried out these four steps in Python, which automates the arithmetic across all the holdings at once. The result is one covariance matrix for each portfolio we study.

Two Matrices, because the Portfolio changed. The Spring 2026 rebalance changed which stocks the Portfolio holds: CVS was sold, while Credo Technology (CRDO) and Rubrik (RBRK) were bought. Since the covariance matrix must contain a row and column for every holding, we computed *two* matrices: a 12×12 matrix for the pre-rebalance portfolio and a 14×14 matrix for the post-rebalance portfolio. The heatmaps below visualize both, with the color scale running from blue (negative covariance) through white (near zero) to red (positive covariance). Because these plot covariances rather than correlations, the boldest cells are driven by the highest-variance holdings; In the post-rebalance map, a single dark cell, the variance of the newly added CRDO, dominates the figure and forces the entire color scale to widen.

Figure 1: Covariance Matrix of the pre-rebalance Portfolio. Diagonal cells are individual variances; Off-diagonal cells are pairwise covariances. The most volatile holding, Garrett Motion (GTX), produces the boldest diagonal cell.

3.2 Reverse-Optimizing the Risk Aversion Constant, γ

We now turn to the risk-aversion constant γ. Rather than assigning an arbitrary value, we infer it from the pre-rebalance Portfolio. The idea, called *reverse-optimization*, is to assume the Northwestern Investment Management Group's Portfolio Committee constructed the pre-rebalance Portfolio optimally and then ask: What value of γ would make that pre-rebalance Portfolio an optimal portfolio? In this way, reverse-engineering the mean-variance optimization turns γ into the one unknown value and lets us solve for it.

Recall from Section 2.3 that an optimal portfolio satisfies the following first-order condition:

$$\mu - \gamma \Sigma w = 0$$

where w is the weight vector, μ is the expected return vector, and Σ is the covariance matrix we computed in Section 3.1. To isolate γ, we multiply both sides by $w^\top$:

$$w^\top(\mu - \gamma\ \Sigma w) = 0$$

Because $\gamma \in R$,

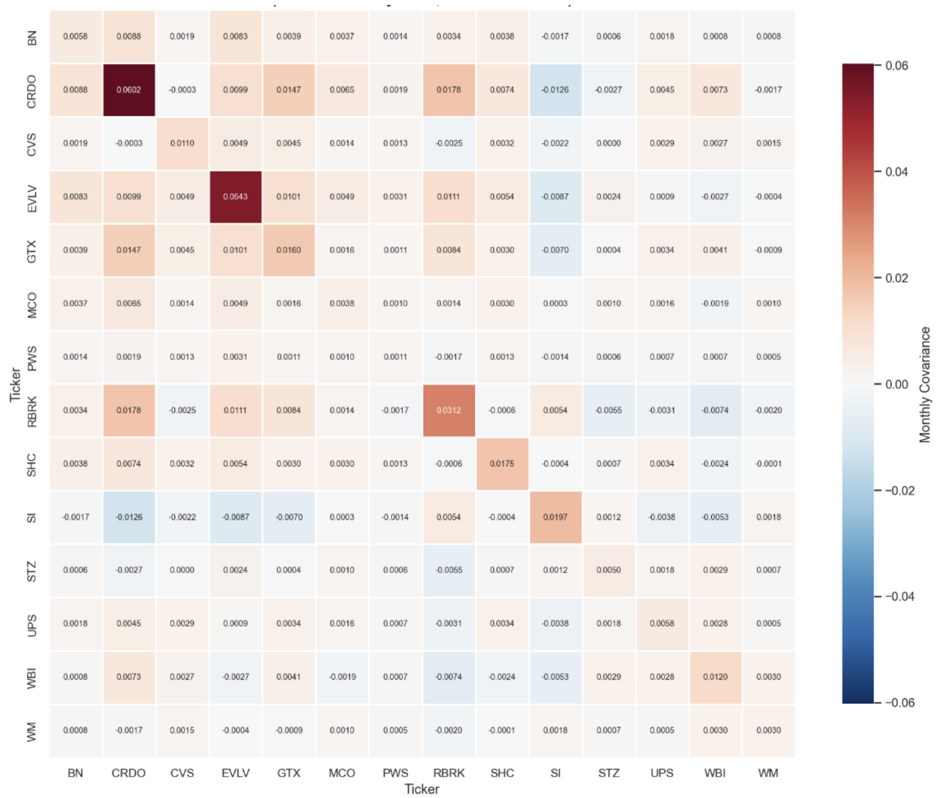

Figure 2: Covariance Matrix of the post-rebalance Portfolio. The diagonal variance of Credo (CRDO), more than four times the next-largest, dominates the map and widens the color scale relative to the pre-rebalance figure.

$$w^\top \mu - \gamma \, w^\top \Sigma w = 0$$

Then, solving for γ gives the following reverse-optimization formula:

$$\boxed{\gamma = \frac{w^\top \mu}{w^\top \Sigma w} = \frac{E[R_p]}{\text{Var}(R_p)}}$$

The formula has a clean meaning. The numerator $w^\top \mu = E[R_p]$ is a portfolio's expected return; The denominator $w^\top \Sigma w = \text{Var}(R_p)$ is a portfolio's variance. Their ratio is how much expected return the investor is demanding for each unit of variance. A risk-averse investor demands a lot of return per unit of risk, giving a large γ; A more risk-tolerant investor accepts less, giving a small γ. In this way, we can effectively interpret γ as the price the investor charges for bearing risk.

A Note on Cash. Both pre-rebalance and post-rebalance Portfolios hold some cash. Cash is assumed to be risk-free, and has zero expected return and zero variance; Cash does not co-move with any stock. Therefore, cash contributes nothing to either the numerator or the denominator, but it still counts toward the total weights, such that holding more cash pulls a portfolio's overall expected return and variance down toward zero.

3.3 The Pre-Rebalance Portfolio

The pre-rebalance Portfolio (with expected returns measured as of March 13, 2026 - the date on which the Northwestern Investment Management Group's Portfolio Committee conducted its meeting for portfolio allocation decisions) consists of twelve stocks and 0.1% in cash. Its weights and expected returns are:

Ticker	μ_i	w_i (%)	Ticker	μ_i	w_i (%)
BN	0.54	13.3	SHC	0.87	8.3
STZ	0.51	2.7	UPS	0.25	2.7
CVS	0.34	10.9	WM	0.36	15.5
GTX	0.93	22.4	EVLV	1.09	4.4
MCO	0.25	8.9	WBI	0.56	3.5
PWS	0.29	4.7	SI	0.41	2.6

Expected Return. Multiplying each weight by its expected return and summing gives:

$$E[R_p] = w^\top \mu = (0.133)(0.54) + (0.027)(0.51) + \cdots + (0.026)(0.41) = 0.5798$$

The pre-rebalance Portfolio's expected return is approximately 58%, lifted heavily by GTX, which carries the largest weight (22.4%) at a high expected return (93%), and by EVLV at 109% expected return.

Portfolio Variance. Feeding the weight vector and the 12×12 covariance matrix into the quadratic form $w^\top \Sigma w$ gives:

$$\text{Var}(R_p) = w^\top \Sigma w = 0.003769, \qquad \sigma_p = \sqrt{0.003769} = 0.0614$$

such that the pre-rebalance Portfolio's monthly volatility is approximately 6.1%

Risk Aversion Constant. Dividing the expected return by the portfolio variance for the risk aversion constant gives,

$$\boxed{\gamma_{\text{pre}} = \frac{0.5798}{0.003769} \approx 153.8}$$

3.4 The Post-Rebalance Portfolio

The post-rebalance Portfolio (with expected returns measured as of May 30, 2026) reflects the following trades: CVS is sold, CRDO and RBRK are added, and Cash has been raised sharply to 16.7%. Its weights and expected returns are:

Ticker	μ_i	w_i (%)	Ticker	μ_i	w_i (%)
BN	0.30	8.1	WM	0.54	8.0
STZ	0.63	1.4	EVLV	0.66	12.5
GTX	0.04	21.7	WBI	0.35	2.2
MCO	0.19	5.1	SI	0.43	1.5
PWS	0.07	3.0	CRDO	0.57	5.4
SHC	0.60	4.8	RBRK	0.08	8.0
UPS	0.14	1.6	Cash	0.00	16.7

Expected Return. With Cash contributing zero,

$$E[R_p] = w^\top \mu = (0.081)(0.30) + (0.014)(0.63) + \cdots + (0.080)(0.08) + (0.167)(0) = 0.2617$$

The expected return has dropped to approximately 26%, less than half the pre-rebalance level. The main reason is GTX: Although it is still the largest position at 21.7%, its remaining upside to the price target shrank from 93% to 4% after the stock price rose 86.24% between March 13, 2026 and May 30, 2026. The 16.7% Cash position also dilutes the expected return.

Portfolio Variance. Using the 14×14 covariance matrix,

$$\mathrm{Var}(R_p) = w^\top \Sigma w = 0.005980, \qquad \sigma_p = \sqrt{0.005980} = 0.0773$$

Even with the large Cash cushion, monthly volatility actually *rose* to approximately 7.7%. The culprit is CRDO, whose monthly variance (0.134) is roughly five times that of any other holding. Furthermore, CRDO exhibits a strong positive covariance with the Portfolio's already-large GTX position, amplifying the overall portfolio variance.

Risk Aversion Constant. Dividing the expected return by the portfolio variance for the risk aversion constant gives,

$$\boxed{\gamma_{\text{post}} = \frac{0.2617}{0.005980} \approx 43.7}$$

3.5 Interpretation of Reverse-Optimized Risk Aversion Constants

Collecting both results:

Portfolio	$E[R_p]$	$\mathrm{Var}(R_p)$	σ_p	γ
Pre-rebalance	0.5798	0.003769	0.0614	153.8
Post-rebalance	0.2617	0.005980	0.0773	43.7

The implied risk aversion constant fell sharply, from about 154 to about 44, a drop of roughly 72%. At first, this seems backwards; The Portfolio Committee raised its Cash holding from almost nothing to 16.7%, which is a risk-reducing decision. However, our result implies the post-rebalance Portfolio is more consistent with a more risk-tolerant investor.

The explanation lies in what γ actually measures. It is the amount of return demanded for each unit of risk. Two things moved together to push it down. First, the numerator collapsed. Expected return fell from 58% to 26% as the Portfolio's winners, GTX above all, rallied toward their price targets and reduced their upsides, while staying as relatively large positions. Second, the denominator rose. Portfolio variance increased from 0.0038 to 0.0060, because the newly added CRDO brought in more risk than the larger Cash holding removed.

3.6 Mean-Variance Optimized Post-Rebalance Portfolio

In this section, we use the reverse-optimized risk aversion constant from the pre-rebalance Portfolio in Section 3.3 as an input to the mean-variance optimization framework applied to the post-rebalance Portfolio. In other words, we seek to determine the weights that would maximize the risk-adjusted expected return of the post-rebalance Portfolio, while preserving the same level of risk aversion implicitly revealed by the pre-rebalance Portfolio. This analysis is meaningful because the post-rebalance Portfolio reflects a notable shift in the Northwestern Investment Management Group's investment style, away from a traditionally conservative value-oriented approach and toward a more aggressive value-growth approach. Applying the risk aversion constant reverse-optimized from the pre-rebalance Portfolio allows us to isolate the impact of the Portfolio's changing opportunity set, while holding the Portfolio Committee's underlying risk preferences constant.

Optimal Portfolio. Solving for the optimal weight vector, w^*, by using the generalized mean-variance optimization for n-asset portfolio derived in Section 2.3 concentrates the entire portfolio in five assets: STZ, EVLV, SHC, WM, and SI.

Table 1: Mean-Variance Optimized post-rebalance Portfolio

Ticker	μ_i	IMG w_i (%)	MVO w_i^* (%)
BN	0.30	8.1	0.0
STZ	0.63	1.4	35.5
GTX	0.04	21.7	0.0
MCO	0.19	5.1	0.0
PWS	0.07	3.0	0.0
SHC	0.60	4.8	20.7
UPS	0.14	1.6	0.0
WM	0.54	8.0	10.6
EVLV	0.66	12.5	23.8
WBI	0.35	2.2	0.0
SI	0.43	1.5	9.4
CRDO	0.57	5.4	0.0
RBRK	0.08	8.0	0.0
Cash	0.00	16.7	0.0
Total		100.0	100.0

This result highlights a fundamental limitation of mean-variance optimization: Optimal weights are highly sensitive to the expected return assumptions. At the same time, it suggests that if the Portfolio Committee maintained the value-oriented risk preferences implied by the pre-rebalance Portfolio, the post-rebalance Portfolio would be optimally allocated in a much more concentrated manner.

4 Conclusion

This project developed the theory of mean-variance optimization from first principles, beginning with a two-asset portfolio and extending the analysis to a generalized n-asset portfolio. To connect this framework with concepts from vector calculus, we introduced scalar line integrals and used them to calculate the average risk-adjusted expected return across all feasible allocations of a two-asset portfolio. Comparing this average risk-adjusted expected return with that of the mean-variance optimized portfolio illustrated how optimization identifies allocations that outperform an average feasible portfolio allocation.

Then, we applied mean-variance optimization to the Northwestern Investment Management Group's portfolio (the Portfolio). We constructed covariance matrices and reverse-optimized the risk aversion constants implied by both the pre-rebalance and post-rebalance Portfolios. This allowed us to apply the pre-rebalance risk aversion constant to the post-rebalance Portfolio and compute the mean-variance optimized post-rebalance Portfolio, which was significantly more concentrated than the raw post-rebalance Portfolio.

One could further develop this project by considering either the Black-Litterman Model, which seeks to improve mean-variance optimization's fundamental limitation of high sensitivity to the expected return assumptions, or the mean-semivariance optimization, which seeks to modify the covariance matrix to better reflect how investors actually perceive risk in practice.

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